Phase 1: Data Preprocessing and Feature Engineering Strategy for Explainable Pediatric Pneumonia Classification

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***Abstract*—I audited the pediatric chest X-ray dataset before model development and converted the raw archive into a reproducible preprocessing strategy. The source contains 17,568 image-extension paths, of which 5,856 are AppleDouble macOS metadata files and 11,712 decode as radiographs. Exact-byte canonicalization reduces the readable paths to 5,824 unique image contents: 1,579 Normal, 2,760 Bacterial Pneumonia, and 1,485 Viral Pneumonia. The audit also identifies duplicated source trees, within-tree repetition, class-dependent image geometry and storage size, uncertain patient identifiers, and 223 perceptual near-duplicate candidates. These findings directly determine my cleaning, missingness, outlier, encoding, scaling, channel-handling, and leakage-prevention decisions. I retain unusual images for review rather than deleting them statistically, keep audit metadata out of the CNN, and plan five-fold stratified group-aware evaluation with every fitted preprocessing operation restricted to training data. Phase 1 therefore establishes a defensible dataset unit and preprocessing design without claiming clinical validity or patient independence.**

***Index Terms*—data preprocessing, chest X-ray, pneumonia, data audit, missingness, outliers, leakage prevention, shortcut learning**

I. Problem Definition and Phase 1 Scope

The project is a supervised three-class image-classification task with

$$Y \in \{\text{Normal}, \text{Bacterial Pneumonia}, \text{Viral Pneumonia}\}. \quad (1)$$

The images come from the pediatric chest X-ray collection associated with Kermany *et al.* [1] and distributed publicly through the Chest X-Ray Images (Pneumonia) dataset [2]. My objective in Phase 1 is not to train a classifier, but to determine what constitutes a valid observation, quantify data-quality risks, and predefine a leakage-safe preprocessing strategy before implementation.

For an image dataset, the assignment's $N \times P$ notation needs two interpretations. The raw filesystem audit is a $17{,}568 \times 25$ metadata matrix, and the deduplicated canonical audit is $5{,}824 \times 33$. The actual model input X is the decoded image, not those metadata columns. Native images have variable dimensions, so there is no fixed pixel feature count before standardization. Under the planned 224×224 grayscale representation, $P = 50{,}176$ intensity positions; a replicated three-channel tensor has 150,528 positions, although the replicated channels do not contain independent acquired information. The target Y is nominal multiclass.

TABLE I
Verified source integrity and canonicalization results

Measure	Result
Image-extension paths	17,568
Readable image paths	11,712
AppleDouble sidecars	5,856
Unique readable SHA-256 contents	5,824
Excess readable byte copies	5,888
Readable hash groups with label conflicts	0
Readable hashes spanning source splits	0
Canonical audit matrix	$5{,}824 \times 33$

The public repository is linked beside my email. It contains the README, dependency specification, audit code, processed evidence, figures, tests, and Phase 1 strategy documents. Raw images and archives are excluded from Git. The current Linux CI run passes the complete preprocessing test suite at commit 4503161.

II. Dataset Audit and Canonicalization

A. Filesystem integrity and exact duplicates

The initial scan found 17,568 paths ending in JPEG/JPG/PNG. Of these, 11,712 decode successfully and 5,856 are verified AppleDouble files under the macOS packaging tree. They are filesystem metadata, not corrupted X-rays. The archive also contains two byte-identical readable source trees. SHA-256 analysis shows 5,824 unique readable contents and 5,888 excess readable copies. Retaining one source tree alone would still preserve within-tree repetition, so canonicalization must be content-based rather than folder-based.

The deterministic canonical rule keeps one eligible readable non-sidecar path per SHA-256 hash, choosing the path with the fewest relative path components and then case-sensitive lexical order. It gives no priority to the original train, validation, or test folder, and all alternative paths remain recorded for provenance. The canonical count of 5,824 is therefore a provisional count of unique readable byte contents, not a claim of 5,824 independent patients.

TABLE II
MISSINGNESS MECHANISM AND TREATMENT DECISION

Field/condition	Assessment	Phase 1 decision
Patient ID candidate	MAR-like w.r.t. observed filename convention; not proven MAR	Do not impute or fabricate identity; preserve missingness and treat grouping as uncertain.
Successful read/hash error fields	Structural N/A	Leave blank; do not impute.
AppleDouble image measurements	Structural N/A	Exclude filesystem metadata from modeling discovery.
Readable image pixels	No missing-pixel mechanism observed	No pixel imputation.

B. Target distribution and class imbalance

The raw inventory counts (4,749 Normal, 8,340 Bacterial, 4,479 Viral) include sidecars and copied trees and are not independent training observations. After exact-content canonicalization, the class counts are 1,579 Normal, 2,760 Bacterial Pneumonia, and 1,485 Viral Pneumonia. The largest/smallest ratio is 1.8586, which is moderate imbalance rather than an extreme rare-class setting. I therefore do not plan SMOTE on raw image vectors. Instead, class weights will be computed from each active training subset only.

III. VARIABLE INVENTORY AND MISSINGNESS DIAGNOSIS

The canonical manifest contains 33 fields: 11 text/path/hash/identifier fields, 9 nominal categorical fields, 7 discrete quantitative fields, 3 continuous fields, and 3 Boolean fields. There are no ordinal or datetime variables. Width, height, channel count, file size, minimum intensity, maximum intensity, and source-file count are discrete quantitative measurements; aspect ratio, mean intensity, and intensity standard deviation are continuous. The full per-column inventory appears in Appendix A.

The raw $17{,}568 \times 25$ audit contains 96,864 missing cells (22.0546%), whereas the canonical $5{,}824 \times 33$ manifest contains 13,228 missing cells (6.8827%). Literal complete-row counts are zero in both tables because successful reads and hashes intentionally leave the corresponding error-message fields blank. That statistic therefore does not indicate poor image completeness. In the canonical manifest, every target and image measurement is complete; the only substantive missing field is `patient_id_if_parseable` for 1,580 rows. The remaining 11,648 missing cells are structural blanks in `read_error` and `hash_error`.

I do not label the identity mechanism as proven MCAR, MAR, or MNAR because the true patient-generating process is unavailable. For planning purposes, the observed dependence on filename convention is most consistent with an MAR-like mechanism. This is a cautious classification, not an empirical proof of MAR.

IV. CLEANING, OUTLIER, AND QUALITY STRATEGY

A. Distribution shape and reproducible review rules

The six numeric audit diagnostics have different distribution shapes. File size and aspect ratio are strongly right-skewed, width and height are moderately/right-skewed, and pooled mean intensity and intensity SD are closer to symmetric. Q-Q diagnostics are saved in the repository and show central linearity for the intensity summaries but class-dependent tail departures. Consequently, I use the $1.5\times$IQR rule as the primary class-blind review screen for all six diagnostics and report $|Z| > 3$ only as a sensitivity check for mean intensity and intensity SD.

The IQR rules flag 1,070 unique contents because flags overlap across metrics. Importantly, the review burden is class-dependent: 837/1,579 Normal images (53.01%), 147/2,760 Bacterial images (5.33%), and 86/1,485 Viral images (5.79%) have at least one canonical flag. File size alone flags 813 Normal images but only three pneumonia images. This makes automatic outlier deletion scientifically unsafe: it would selectively remove one class based on acquisition/storage characteristics rather than demonstrated image invalidity.

For the near-symmetric intensity summaries, $|Z| > 3$ flags 22 mean-intensity images (0.38%) and 21 intensity-SD images (0.36%). These are also review indicators only. Negative lower IQR fences for height or file size are statistical artifacts of the rule, not physically meaningful values.

B. Imputation and capping applicability

Mean, median, mode, and KNN imputation are not applied to decoded pixels because the readable canonical images do not contain missing pixels. They are also inappropriate for unknown patient identity and structural error fields. Pixel Winsorization is rejected because image-level metadata outliers do not justify changing radiographic intensities. If a separate tabular metadata diagnostic is implemented later, any capping will be fitted on training data only and will modify that diagnostic table, never the X-ray pixels.

V. STATISTICAL ANALYSIS AND FEATURE-ENGINEERING DECISIONS

A. Association tests matched to data type

For numeric diagnostics versus the three-class target, I use Kruskal–Wallis because the distributions are skewed and heteroscedastic. I report the rank-based effect $\eta^2_H = (H - k + 1)/(N - k)$ and mutual information (MI; 10 pooled quantile bins, 499 label permutations, seed 42). For the nominal storage mode, I use Pearson's Chi-square with Cramer's V. Holm correction controls the family of inferential p-values [5]. Because true patient independence is unverified, p-values are interpreted as exploratory; effect sizes and preprocessing implications carry more weight.

The large effects for file size and image geometry show that acquisition/export characteristics are strongly entangled with class. This is a plausible shortcut-learning risk rather than evidence of disease-specific causation [3], [4]. I therefore keep these variables for audit and diagnostic baselines but do not feed them to the CNN.

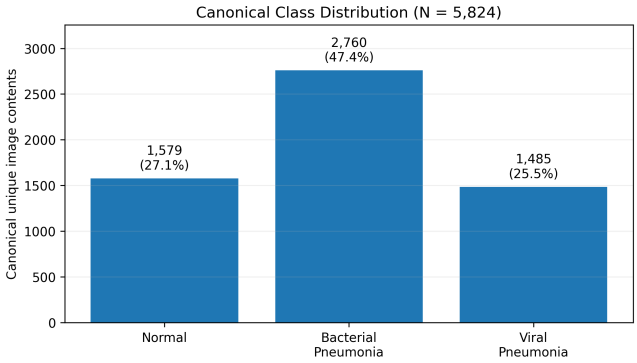


Fig. 1. Canonical target distribution after exact-content deduplication. Counts sum to $N = 5{,}824$.

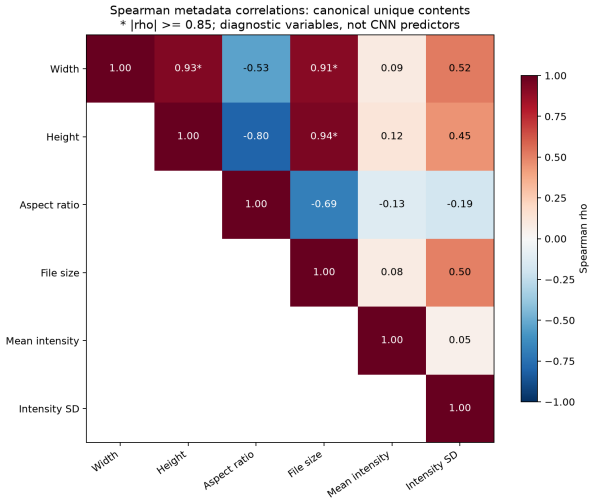


Fig. 2. Spearman correlation among numeric audit diagnostics. Asterisks mark $|\rho| \geq 0.85$. These are diagnostic metadata, not CNN predictors.

TABLE III
CANONICAL OUTLIER-REVIEW RULES AND OBSERVED FLAG COUNTS ($N = 5{,}824$)

Diagnostic	Skewness	Q1	Q3	Lower fence	Upper fence	Flagged
Width	0.5842	1056	1560	300	2316	73
Height	0.9588	688	1187	-60.5	1935.5	127
Aspect ratio	1.2257	1.2621	1.5861	0.7760	2.0722	98
File size (bytes)	2.6401	56,041.75	238,312	-217,363.63	511,717.38	816
Mean intensity	-0.0210	111.5568	134.2946	77.4500	168.4014	105
Intensity SD	-0.2296	50.7322	63.7061	31.2714	83.1669	40

TABLE IV
METADATA-TO-TARGET ASSOCIATION AUDIT. METADATA REMAIN DIAGNOSTIC AND ARE NOT CNN INPUT FEATURES.

Variable	Test statistic	Effect	MI (nats)	Holm-adjusted p	Interpretation
File size	$H = 3311.88$	$\eta^2_H = 0.5686$	0.4517	$< 10^{-300}$	Very strong acquisition/storage association
Height	$H = 2431.91$	$\eta^2_H = 0.4174$	0.2688	$< 10^{-300}$	Strong geometry association
Width	$H = 2229.50$	$\eta^2_H = 0.3827$	0.2380	$< 10^{-300}$	Strong geometry association
Aspect ratio	$H = 1614.60$	$\eta^2_H = 0.2770$	0.1584	$< 10^{-300}$	Moderate geometry association
Intensity SD	$H = 993.80$	$\eta^2_H = 0.1704$	0.1335	6.33×10^{-216}	Moderate contrast association
Mean intensity	$H = 51.21$	$\eta^2_H = 0.0085$	0.0248	7.57×10^{-12}	Statistically detectable, practically small
Image mode	$\chi^2 = 148.30$	$V = 0.1596$	0.0181	1.88×10^{-32}	Small/moderate storage-mode association

B. Correlation, VIF, and deterministic collinearity handling

Spearman correlation is used because several diagnostics are non-normal and monotonic rather than linearly related. The strongest pairs are width–height ($\rho = 0.9296$), width–file size ($\rho = 0.9128$), and height–file size ($\rho = 0.9383$), as shown in Fig. 2. Raw-scale VIF is 15.50 for width and 30.43 for height, exceeding the diagnostic threshold of 10; aspect ratio is 4.33, file size 3.31, mean intensity 1.03, and intensity SD 1.32. VIF is treated as a complementary linear-collinearity diagnostic rather than a substitute for rank correlation [6].

No pixels are removed because of metadata collinearity. For a separate metadata-only diagnostic baseline, the removal rule is deterministic: remove a structurally derived collinear variable before its source measurements (aspect ratio first when implicated), recompute VIF, then repeatedly remove the highest-VIF variable above 10; ties are broken by weaker target MI and then alphabetically. This rule is restricted to the tabular diagnostic branch.

C. Target encoding

The target is nominal. I encode Normal$\rightarrow$ 0, Bacterial Pneumonia$\rightarrow$ 1, and Viral Pneumonia$\rightarrow$ 2 as integer class indices for multiclass loss functions. The integers are labels only and do not imply an ordinal relationship. No one-hot encoding of Y is required for standard multiclass cross-entropy.

VI. IMAGE TRANSFORMATION AND SCALING STRATEGY

A. Geometry and channels

I standardize images to a 224×224 canvas using aspect-preserving resize and symmetric padding. The resize scale is $\min(224/w, 224/h)$ with antialiased bilinear interpolation; if an odd padding pixel is required, it is placed on the bottom or

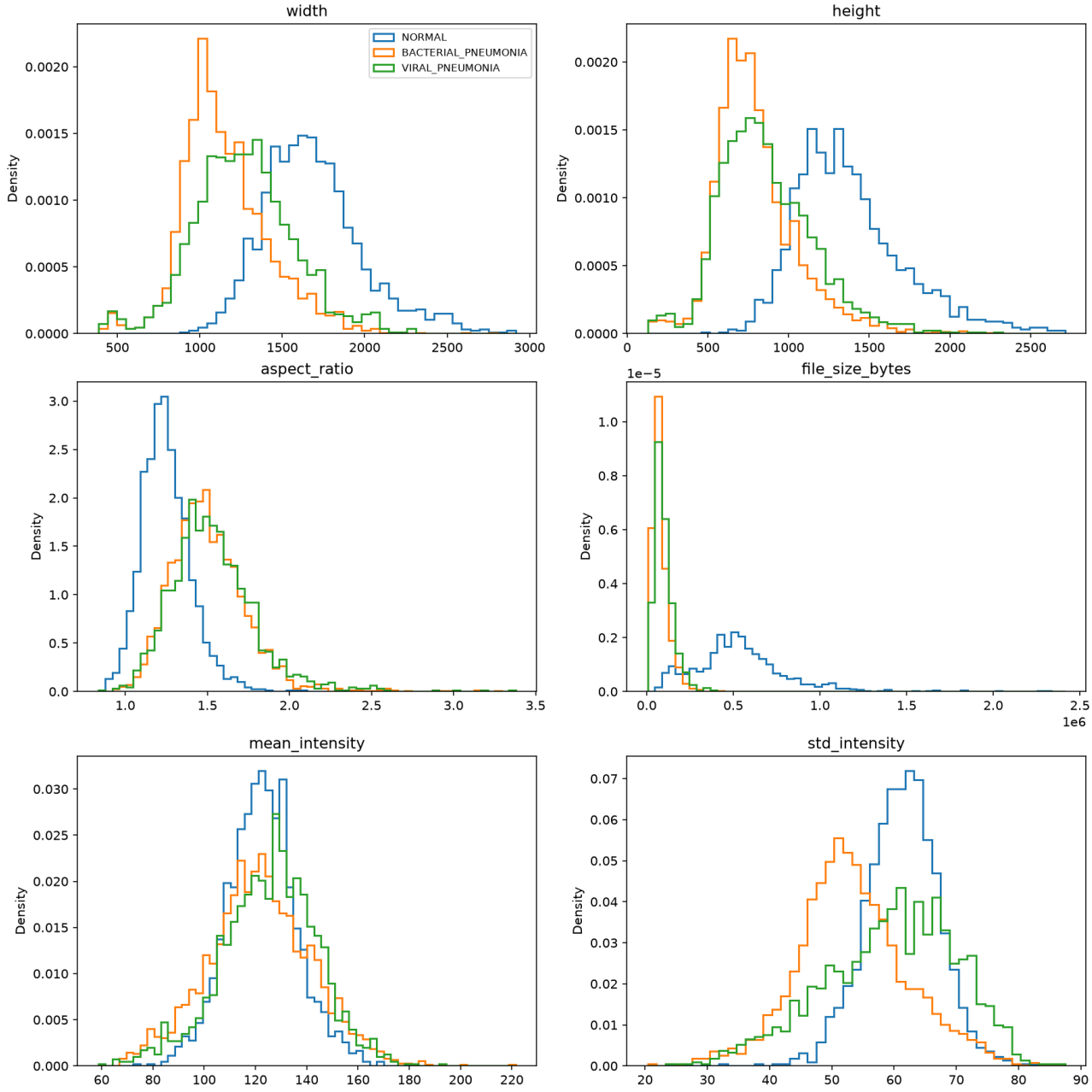


Fig. 3. Class-wise distributions of the six numeric audit diagnostics. Geometry and file size differ substantially by class, while mean intensity has much smaller practical separation. These patterns motivate robust statistics and shortcut-risk analysis.

right. Direct square stretching is not the primary choice because canonical aspect ratios span approximately 0.835–3.379, and routine cropping risks removing peripheral anatomy.

The canonical set contains 5,541 one-channel `L` images and 283 RGB images. All 283 RGB images have pixel-identical R, G, and B channels, so one grayscale intensity channel is sufficient. For pretrained three-channel networks, that channel can be replicated three times. Channel standardization does not remove text, laterality markers, borders, or other possible shortcuts, so these remain explicit review concerns.

B. Scaling, normalization, and augmentation

For a custom CNN baseline, pixels are converted to `float32` and divided by 255. This is a fixed unit conversion, not a statistic fitted to the dataset. If pretrained weights are used, the replicated grayscale tensor will follow the selected weights' documented normalization. Any dataset-derived mean/standard deviation must be estimated only on the current training subset and then frozen for validation/evaluation.

The initial training-only augmentation set is intentionally conservative: rotations within $\pm 5^\circ$ and translations up to 2% of the canvas, with visual checks to prevent anatomy loss. I do not

plan vertical flips, 90/180-degree rotations, aggressive crops, aspect-ratio stretching, strong brightness/contrast shifts, color jitter, or unvalidated erasing. These transformations are not assumed clinically invariant merely because they are common computer-vision defaults.

Padding itself can preserve class-associated aspect-ratio information. I therefore treat geometry choice as a shortcut-risk decision and will compare plausible geometry alternatives using internal validation rather than selecting one from outer-evaluation performance. Passing such checks will not be interpreted as proof of clinical validity.

VII. Near-Duplicates, Identity Uncertainty, and Leakage Risk

Exact SHA-256 deduplication removes byte-identical repetition but not re-encoded or visually similar images. I therefore apply a 64-bit perceptual hash: grayscale conversion, 32×32 resize, orthonormal 2-D DCT-II, upper-left 8×8 coefficients, median thresholding, and BK-tree search at Hamming radius ≤ 4 [7]. The screen finds 223 candidate pairs involving 262 unique images: 175 same-class, 48 cross-class, 27 crossing original source splits, and 10 satisfying both conditions. Sixteen pairs occur at distance 2 and 207 at distance 4. These are review candidates, not confirmed duplicate examinations or patients.

Filename evidence is also useful but uncertain. The canonical set contains 1,443 Normal relation groups, 1,437 bacterial-class groups, and 1,216 viral-class groups. The bacterial and viral counts overlap because 979 numeric tokens appear in both subtype namespaces: $1437 + 1216 - 979 = 1674$ merged pneumonia groups; adding 1,443 Normal groups gives 3,117 filename-derived base groups. Cross-subtype merging is deliberately conservative and is not presented as proof of shared patient identity.

For each pHash candidate edge, a confirmed relation is merged into the leakage-control graph, a rejected edge is omitted, and an unresolved edge is retained only as a split constraint so that its endpoints cannot cross training and evaluation roles. Thus 3,117 is the filename-only base count rather than a guaranteed final component count.

VIII. Partitioning and Leakage Prevention

The original source split is not reused: across the discovered inventory it is 15,648 train, 48 validation, and 1,872 test paths, and the validation folder corresponds to only 16 readable source images per tree. Instead, after the canonical membership and relation constraints are frozen, the primary strategy is

```
StratifiedGroupKFold(n_splits=5,
 shuffle=True, random_state=42).
```

Five folds are feasible because the smallest class contains 1,485 images and at least 1,216 filename-derived relation groups before additional candidate-edge merging.

Each outer fold uses approximately 80% development and 20% evaluation. Within the development portion, I reserve approximately 10% group-aware for internal validation, giving about 72% training ($\sim 4{,}193$ images), 8% validation (~ 466), and 20% outer evaluation ($\sim 1{,}165$) before group-size rounding. Exact counts can differ because keeping a relation component intact takes priority over exact percentages. Temporal splitting is not applicable because trustworthy acquisition timestamps are unavailable.

Leakage-safe order: freeze canonical contents and relation constraints → create outer group-aware development/evaluation split → fit any learned preprocessing, class weights, and augmentation on training only → use internal validation for preprocessing/model decisions → lock the configuration → evaluate once on the untouched outer fold.

Fig. 4. Planned isolation of fitted preprocessing from evaluation data. Whole-cohort Phase 1 statistics are descriptive audit evidence, not fitted transformation state.

Fixed division by 255 and externally published pretrained normalization constants do not learn from this dataset. Every dataset-derived statistic, capping rule, normalization parameter, class weight, or learned transform must be recomputed from the active training subset. This is the central leakage-control rule for Phase 2 implementation.

IX. Phase 1 Decision Summary and Limitations

The principal limitation is that byte uniqueness and filename grouping do not establish patient independence. The labels are dataset annotations rather than microbiological confirmation, and the pediatric source population does not justify direct clinical claims for other populations or settings. The metadata associations also show that acquisition/storage shortcuts are plausible. Phase 1 therefore defines a defensible engineering strategy, not a clinically validated diagnostic system.

The dataset is ready to move into implementation once manual review status is recorded and the final canonical membership is frozen. The strongest conclusion from this phase is not a model result; it is that careful canonicalization, explicit missingness treatment, robust outlier review, conservative image transformations, and group-aware leakage control are necessary before any reported performance can be trusted.

Appendix A
Complete Metadata Inventory and Baseline Missingness

Table VI lists every canonical metadata field, its semantic type, and baseline missingness in the original and canonical audit matrices. Fields added only during canonicalization are marked "–" for the original audit. The target is `target_label`; all other fields are audit/provenance variables rather than CNN predictors.

For the original audit, overall missingness is 96,864/439,200 cells (22.0546%); complete rows are 0 because optional error columns are structurally blank, while 8,544 rows are complete when those two error fields are excluded. For the canonical manifest, overall missingness is 13,228/192,192 cells (6.8827%); again, literal complete rows are 0 because the two error fields are blank on every successful row. Excluding the two optional error fields, 4,244 rows are complete; the remaining 1,580 rows differ only by the unverified patient-ID candidate field.

TABLE V
FINAL PHASE 1 DECISIONS DERIVED FROM THE AUDIT

Decision area	Evidence	Final strategy
Observation unit	11,712 readable paths reduce to 5,824 unique SHA-256 contents	Use the canonical manifest; preserve alternative paths as provenance.
Filesystem metadata	5,856 verified AppleDouble files	Exclude from modeling discovery; retain raw archive unchanged.
Missingness	Canonical image measurements complete; ID candidates missing for 1,580 rows	No pixel/ID/error-field imputation.
Outliers	1,070 canonical contents flagged; flagging strongly class-dependent	Review only; no automatic deletion or pixel Winsorization.
Class imbalance	1,579/2,760/1,485	Use training-fold class weights; no SMOTE on raw images.
Channels	283 RGB images have identical channels	Common grayscale representation; replicate for 3-channel pretrained input.
Geometry	Wide size/aspect-ratio range; aspect ratio associated with target	224×224 aspect-preserving resize + padding; monitor shortcut risk.
Metadata associations	Large effects for file size/dimensions	Treat as diagnostics, not CNN predictors.
Duplicates/relations	223 pHash candidates; uncertain filename identities	Use conservative relation constraints and group-aware splitting.
Leakage	Fitted preprocessing can contaminate validation/evaluation	Fit every learned operation on training only.

TABLE VI
COMPLETE VARIABLE INVENTORY AND PER-COLUMN BASELINE MISSINGNESS

#	Field	Semantic type	Original missing	Canonical missing
1	filepath	Text/path	0 (0.00%)	0 (0.00%)
2	filename	Text	0 (0.00%)	0 (0.00%)
3	original_split	Nominal	0 (0.00%)	0 (0.00%)
4	original_folder	Nominal	0 (0.00%)	0 (0.00%)
5	target_label	Nominal target	0 (0.00%)	0 (0.00%)
6	patient_id_if_parseable	Identifier	9,024 (51.37%)	1,580 (27.13%)
7	patient_id_status	Nominal	0 (0.00%)	0 (0.00%)
8	label_parse_status	Nominal	0 (0.00%)	0 (0.00%)
9	label_parse_reason	Nominal	0 (0.00%)	0 (0.00%)
10	width	Discrete	5,856 (33.33%)	0 (0.00%)
11	height	Discrete	5,856 (33.33%)	0 (0.00%)
12	aspect_ratio	Continuous	5,856 (33.33%)	0 (0.00%)
13	image_mode	Nominal	5,856 (33.33%)	0 (0.00%)
14	channel_count	Discrete	5,856 (33.33%)	0 (0.00%)
15	file_size_bytes	Discrete	0 (0.00%)	0 (0.00%)
16	mean_intensity	Continuous	5,856 (33.33%)	0 (0.00%)
17	std_intensity	Continuous	5,856 (33.33%)	0 (0.00%)
18	min_intensity	Discrete	5,856 (33.33%)	0 (0.00%)
19	max_intensity	Discrete	5,856 (33.33%)	0 (0.00%)
20	readable	Boolean	0 (0.00%)	0 (0.00%)
21	sha256_hash	Text/hash	0 (0.00%)	0 (0.00%)
22	intensity_basis	Nominal	5,856 (33.33%)	0 (0.00%)
23	read_error	Text/error	11,712 (66.67%)	5,824 (100%)
24	hash_error	Text/error	17,568 (100%)	5,824 (100%)
25	is_macos_sidecar	Boolean	0 (0.00%)	0 (0.00%)
26	all_filepaths	Text/path	– (–)	0 (0.00%)
27	all_original_splits	Text	– (–)	0 (0.00%)
28	all_target_labels	Text	– (–)	0 (0.00%)
29	all_filenames	Text	– (–)	0 (0.00%)
30	source_file_count	Discrete	– (–)	0 (0.00%)
31	alternative_filepaths	Text/path	– (–)	0 (0.00%)
32	manifest_status	Nominal	– (–)	0 (0.00%)
33	patient_grouping_validated	Boolean	– (–)	0 (0.00%)

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